T3000 API & Protocol

Message: UPDATE_ENTRY · built Nov 13, 2026

Message: UPDATE_ENTRY

UPDATE_ENTRY updates a single field of a specific entry. This is the fastest way to change one value without sending a full entry update.

Common Uses:

- Quick value changes
- Toggle on/off states
- Single setpoint adjustments
- Manual/auto mode switches

Overview

Message Type: UPDATE_ENTRY **File Location:** `BacnetWebView.cpp` line 2167 **Purpose:** Update single entry field

Request Format

JSON

```
{
  "action": "UPDATE_ENTRY",
  "source": 1,
  "panelId": 0,
  "serialNumber": 12345,
  "entryType": 2,
  "index": 5,
  "field": "value",
  "value": 500,
  "objectinstance": 123456
}
```

Parameters

Parameter	Type	Required	Description
`action`	String	✅	Must be "UPDATE_ENTRY"
`source`	Integer	✅	Message source: 0=T3000, 1=WebUI
`panelId`	Integer	✅	Panel number (0-254)
`serialNumber`	Integer	✅	Device serial number
`entryType`	Integer	✅	Type of entry
`index`	Integer	✅	Entry index
`field`	String	✅	Field name to update
`value`	Any	✅	New value
`objectinstance`	Integer	✅	BACnet object instance

Response Format

JSON

```
{
  "action": "UPDATE_ENTRY_RES",
  "data": {
    "status": true
  }
}
```

Updatable Fields by Type

Outputs (BAC_OUT)

- `value` - Output value (0-1000)
- `auto_manual` - 0=Auto, 1=Manual
- `range` - Unit selection

Variables (BAC_VAR)

- `value` - Variable value
- `auto_manual` - 0=Auto, 1=Manual

Programs (BAC_PRG)

- `on_off` - 0=Off, 1=On
- `auto_manual` - 0=Auto, 1=Manual

Controllers (BAC_PID)

- `setpoint` - Setpoint value
- `auto_manual` - 0=Auto, 1=Manual
- `action` - 0=Direct, 1=Reverse
- `proportional` - P gain
- `reset` - I gain
- `rate` - D gain

Implementation

C++

```
case WEBVIEW_MESSAGE_TYPE::UPDATE_ENTRY:
{
    int entry_type = json.get("entryType", -1).asInt();
    int index = json.get("index", -1).asInt();
    std::string field = json.get("field", "").asString();

    switch (entry_type)
    {
        case BAC_OUT:
            if (field == "value")
            {
                int value = json.get("value", 0).asInt();
                g_Output_data[panel_id].at(index).value = value;
                WriteOutputValue(objectinstance, index, value);
            }
            break;
        // ... other types ...
    }

    tempjson["action"] = "UPDATE_ENTRY_RES";
    tempjson["data"]["status"] = true;
    break;
}
```

Performance

Operation	Time
Value update	100-200ms
Mode change	100-200ms
PID parameter	150-300ms

Usage Examples

Example 1: Change Output Value

JSON

```
{
  "action": "UPDATE_ENTRY",
  "panelId": 0,
  "serialNumber": 12345,
  "entryType": 2,
  "index": 0,
  "field": "value",
  "value": 750,
  "objectinstance": 123456
}
```

Example 2: Toggle Program

JSON

```
{
  "action": "UPDATE_ENTRY",
  "panelId": 0,
  "serialNumber": 12345,
  "entryType": 6,
  "index": 3,
  "field": "on_off",
  "value": 1,
  "objectinstance": 123456
}
```

Example 3: Change PID Setpoint

JSON

```
{
  "action": "UPDATE_ENTRY",
  "panelId": 0,
  "serialNumber": 12345,
  "entryType": 11,
  "index": 0,
  "field": "setpoint",
  "value": 720,
  "objectinstance": 123456
}
```

Error Handling

JSON

```
{
  "action": "ERROR_RES",
  "error": {
    "code": -2,
    "message": "Invalid field",
    "details": "Field 'xyz' is not writable for entry type 2"
  }
}
```

See Also

- UPDATE_WEBVIEW_LIST - Bulk updates
- GET_WEBVIEW_LIST - Read values